

Training-Free Token Reduction on MotionBench

Results

HuVLLM Project

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Eleven training-free video-LLM token-reduction methods, evaluated on MotionBench (8,052 questions, 4,018 scoreable) across two backbones — LLaVA-OV-7B and Qwen3-VL-8B — at 32 frames. Results only: full methodology, the verification gate, and limitations are in `paper.tex`. At $n = 4,018$ the binomial 95% confidence half-width is ± 1.54 points; * marks a difference significant by McNemar’s test on paired predictions ($\chi^2 \geq 3.84$, $p < 0.05$).

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1 Common settings

Held fixed across every method and both backbones:

- **32 frames**, sampled uniformly per video.
- **Greedy decoding** (`do_sample=False`, `max_new_tokens=16`), batch size 1.
- **Letter-match scoring** (A–D) against ground truth; the 4,034 NA items are excluded, leaving 4,018 scoreable.
- Methods exposing a single retention knob are standardized to **15% nominal retention**; methods with no such knob (frame-selection methods, or methods whose budget is fixed internally) keep their published defaults — forcing 15% on those would break paper fidelity.

Per-method settings, mechanisms and measured token budgets are in Table 4 (§5).

2 Master table

Table 1 is every accuracy figure in this report in one place: both backbones, all six subcategories, both baselines, and every retention level actually run. **Only 10%, 15% and 25% nominal retention exist in the underlying results cache** — no 20% sweep was run for any method, so it is not shown; adding it would require new evaluation jobs, not a different table. Four of eleven methods (FastV, FlashVID, HoliTom, PruneVID) expose a comparable retention knob and appear at all three levels; the remaining seven appear once, at their standardized or default setting (§1). [§] AIM’s released code ships two active merge steps, so its LLaVA-OV cell runs at 25% retention, not 15%. Rounding is half away from zero throughout (see note, Table 1 caption).

Table 1: Master results table. AO = Action Order, CM = Camera Motion, LM = Location-related Motion, MR = Motion Recognition, MO = Motion-related Objects, RC = Repetition Count (% correct within category, chance = 25%). Δ is vs. that row’s backbone baseline. * = significant. Six Repetition Count cells at the 15% nominal setting land on an exact x.25 rounding tie (e.g. 101/400); this table rounds half away from zero, matching `paper.tex` — naive recomputation with default rounding disagrees on these cells by 0.1 point.

Method	Backbone	Retention	Overall	Δ	AO	CM	LM	MR	MO	RC
<i>Baseline</i>	LLaVA-OV-7B	—	<i>52.66</i>	—	40.5	45.2	55.5	57.0	71.2	23.8
<i>Baseline</i>	Qwen3-VL-8B	—	<i>62.52</i>	—	46.1	63.1	65.0	67.3	79.0	33.8
<i>LLaVA-OV-7B, standardized setting</i>										
DyCoke	LLaVA-OV-7B	15%	53.36	+0.70	38.9	48.8	55.7	58.2	70.9	25.3
FlashVID	LLaVA-OV-7B	15%	53.31	+0.65	39.9	45.7	53.8	58.0	71.7	28.2
HoliTom	LLaVA-OV-7B	15%	53.14	+0.47	40.8	49.6	52.6	57.0	71.4	27.3
MDP3	LLaVA-OV-7B	15%	53.06	+0.40	40.5	49.6	53.5	56.8	71.6	26.5
VideoITG	LLaVA-OV-7B	15%	52.86	+0.20	40.1	47.0	53.7	57.6	70.1	26.8
AIM	LLaVA-OV-7B	25% [§]	52.86	+0.20	41.4	48.1	54.2	57.0	71.9	22.3
STTM	LLaVA-OV-7B	15%	51.72	−0.95	39.9	48.1	53.8	53.5	70.6	28.8
PruneVID-OV	LLaVA-OV-7B	15%	38.20	−14.46*	32.0	34.8	35.5	38.4	52.9	27.3
FastV	LLaVA-OV-7B	15%	36.78	−15.88*	32.8	31.9	34.1	35.7	53.8	25.3
<i>Qwen3-VL-8B, standardized setting</i>										
PruneVID	Qwen3-VL-8B	15%	62.17	−0.35	46.1	62.9	64.1	67.1	79.1	32.5
DyCoke	Qwen3-VL-8B	15%	61.85	−0.67	45.1	62.6	63.9	66.4	78.4	34.5
HoliTom	Qwen3-VL-8B	15%	60.33	−2.19*	44.7	61.3	61.0	66.4	76.2	29.0
MDP3	Qwen3-VL-8B	15%	59.66	−2.86*	44.9	64.7	62.8	64.0	77.4	23.0
FastV	Qwen3-VL-8B	15%	59.01	−3.51*	46.2	61.0	61.9	63.5	72.2	30.5
VisionZip (contextual)	Qwen3-VL-8B	15%	58.81	−3.71*	43.2	57.4	61.9	64.2	74.3	29.5
STTM	Qwen3-VL-8B	15%	57.07	−5.45*	44.3	60.0	61.0	60.6	73.8	23.8
FlashVID	Qwen3-VL-8B	15%	56.65	−5.87*	43.4	57.9	57.1	61.2	73.0	27.0
VideoITG	Qwen3-VL-8B	15%	56.35	−6.17*	45.3	53.5	59.3	60.2	75.4	22.2
AIM	Qwen3-VL-8B	15%	55.97	−6.55*	45.9	56.6	59.5	58.3	72.2	27.0
<i>LLaVA-OV-7B, retention sweep (10% and 25%; 15% already listed above)</i>										
FastV	LLaVA-OV-7B	10%	36.06	−16.60*	32.9	31.9	33.9	34.1	53.3	24.5
FastV	LLaVA-OV-7B	25%	36.73	−15.93*	32.0	31.7	33.3	36.3	53.0	26.0

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Table 1 (continued)

Method	Backbone	Retention	Overall	Δ	AO	CM	LM	MR	MO	RC
FlashVID	LLaVA-OV-7B	10%	52.51	-0.15	41.0	46.8	53.3	56.3	70.7	26.5
FlashVID	LLaVA-OV-7B	25%	53.36	+0.70	42.2	47.5	55.1	57.5	71.9	23.8
HoliTom	LLaVA-OV-7B	10%	52.76	+0.10	40.5	48.8	53.3	55.8	71.2	29.0
HoliTom	LLaVA-OV-7B	25%	53.41	+0.75	40.7	48.3	54.8	57.7	72.3	24.5
PruneVID-OV	LLaVA-OV-7B	10%	38.10	-14.56*	33.5	33.5	36.8	38.2	52.3	25.2
PruneVID-OV	LLaVA-OV-7B	25%	38.38	-14.29*	32.2	33.2	36.6	38.8	53.5	26.2
<i>Qwen3-VL-8B, retention sweep (10% and 25%; 15% already listed above)</i>										
FastV	Qwen3-VL-8B	10%	56.92	-5.60*	44.5	60.5	59.0	61.4	69.9	28.0
FastV	Qwen3-VL-8B	25%	60.60	-1.92*	46.4	62.3	63.7	65.2	74.1	32.8
FlashVID	Qwen3-VL-8B	10%	54.73	-7.79*	44.5	54.5	56.0	58.4	70.9	25.0
FlashVID	Qwen3-VL-8B	25%	59.36	-3.16*	43.9	62.3	61.0	63.8	75.7	29.8
HoliTom	Qwen3-VL-8B	10%	60.05	-2.46*	44.9	60.8	59.7	65.7	77.4	28.7
HoliTom	Qwen3-VL-8B	25%	60.43	-2.09*	45.1	61.0	61.0	66.0	77.5	28.7
PruneVID	Qwen3-VL-8B	10%	60.23	-2.29*	46.6	60.8	61.0	65.2	77.1	28.7
PruneVID	Qwen3-VL-8B	25%	61.40	-1.12*	46.1	60.0	63.0	66.6	78.7	31.5

3 Results by backbone

3.1 LLaVA-OV-7B

Table 2: LLaVA-OV-7B, 32 frames, 15% nominal retention except where noted. All nine methods that run on this backbone. [§] AIM runs at 25% retention (two active merge steps ship in the released code).

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
DyCoke	53.36	+0.70	38.9	48.8	55.7	58.2	70.9	25.3
FlashVID	53.31	+0.65	39.9	45.7	53.8	58.0	71.7	28.2
HoliTom	53.14	+0.47	40.8	49.6	52.6	57.0	71.4	27.3
MDP3	53.06	+0.40	40.5	49.6	53.5	56.8	71.6	26.5
VideoITG	52.86	+0.20	40.1	47.0	53.7	57.6	70.1	26.8
AIM [§]	52.86	+0.20	41.4	48.1	54.2	57.0	71.9	22.3
<i>Baseline</i>	<i>52.66</i>	—	<i>40.5</i>	<i>45.2</i>	<i>55.5</i>	<i>57.0</i>	<i>71.2</i>	<i>23.8</i>
STTM	51.72	−0.95	39.9	48.1	53.8	53.5	70.6	28.8
PruneVID-OV*	38.20	−14.46	32.0	34.8	35.5	38.4	52.9	27.3
FastV*	36.78	−15.88	32.8	31.9	34.1	35.7	53.8	25.3

3.2 Qwen3-VL-8B

Table 3: Qwen3-VL-8B, 32 frames, 15% nominal retention. All ten portable methods. VisionZip is the contextual half only — Qwen3-VL’s vision tower has no CLS token, so the dominant half is not implementable; not the published method. [§] marks a configuration the method’s authors do not publish for this backbone.

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
<i>Baseline</i>	<i>62.52</i>	—	<i>46.1</i>	<i>63.1</i>	<i>65.0</i>	<i>67.3</i>	<i>79.0</i>	<i>33.8</i>
PruneVID	62.17	−0.35	46.1	62.9	64.1	67.1	79.1	32.5
DyCoke	61.85	−0.67	45.1	62.6	63.9	66.4	78.4	34.5
HoliTom*	60.33	−2.19	44.7	61.3	61.0	66.4	76.2	29.0
MDP3*	59.66	−2.86	44.9	64.7	62.8	64.0	77.4	23.0
FastV*	59.01	−3.51	46.2	61.0	61.9	63.5	72.2	30.5
VisionZip (ctx.)*	58.81	−3.71	43.2	57.4	61.9	64.2	74.3	29.5
STTM* [§]	57.07	−5.45	44.3	60.0	61.0	60.6	73.8	23.8
FlashVID* [§]	56.65	−5.87	43.4	57.9	57.1	61.2	73.0	27.0
VideoITG*	56.35	−6.17	45.3	53.5	59.3	60.2	75.4	22.2
AIM*	55.97	−6.55	45.9	56.6	59.5	58.3	72.2	27.0

4 Figures

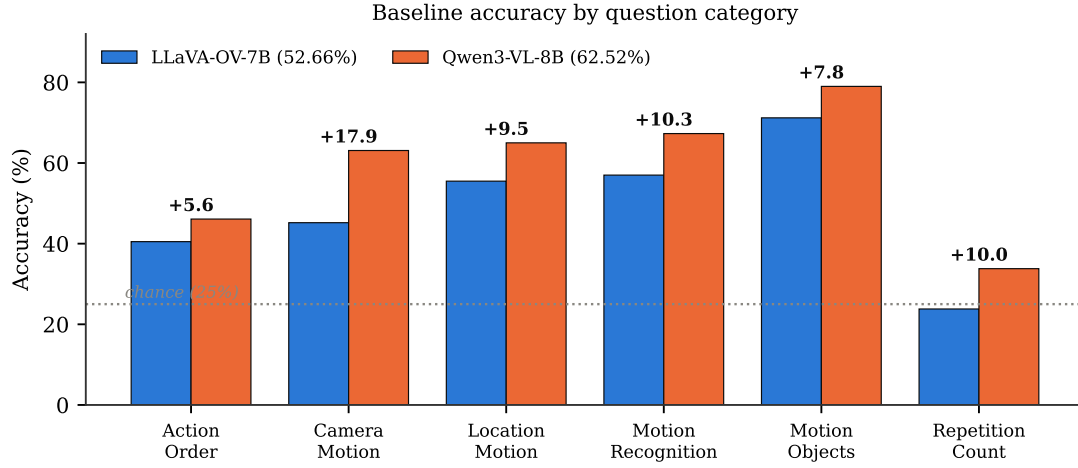


Figure 1: Baseline accuracy by category, both backbones (bar).

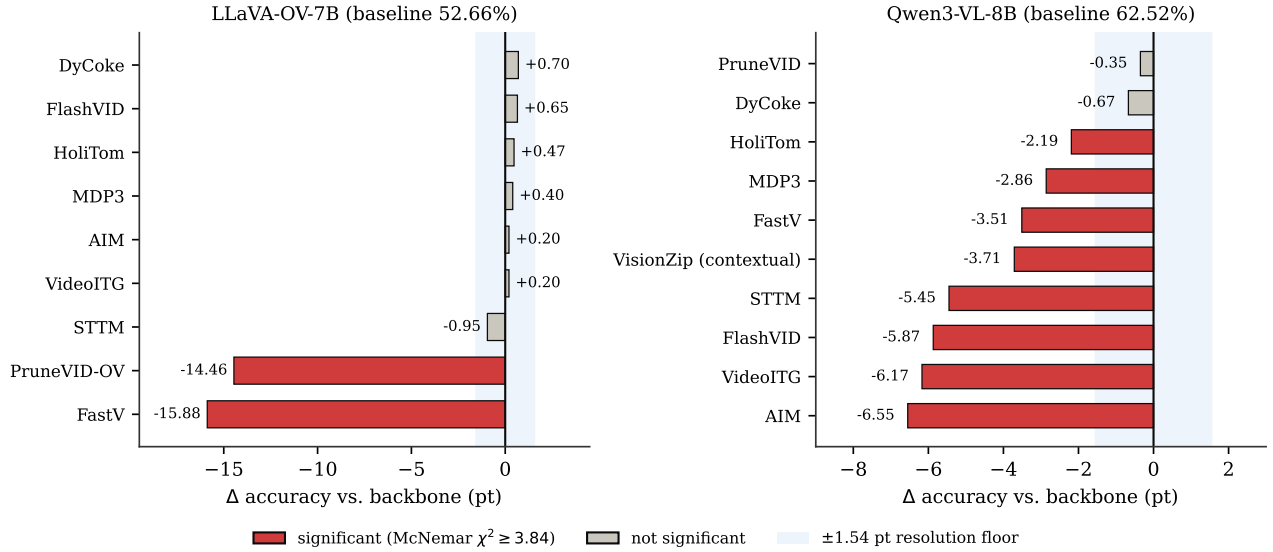


Figure 2: Change vs. backbone, every method, both backbones (bar). Red = significant by McNemar’s test; the shaded band is the ± 1.54 -point resolution floor.

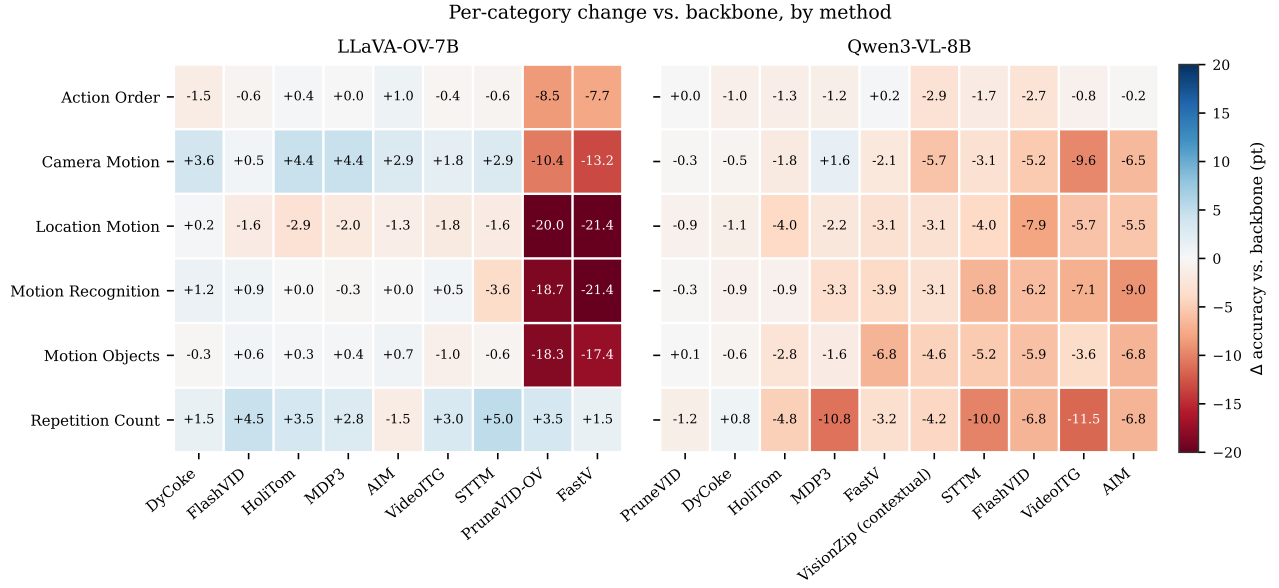


Figure 3: Per-category change vs. backbone, by method (heatmap). Blue = gain, red = loss.

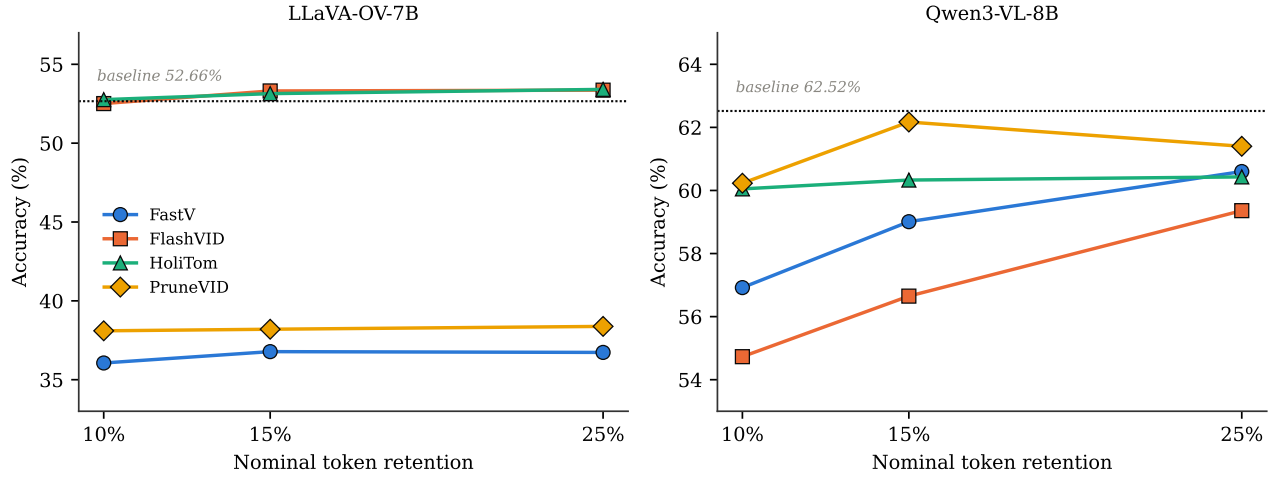


Figure 4: Accuracy against nominal token retention, the four methods with a retention knob (dot + line).

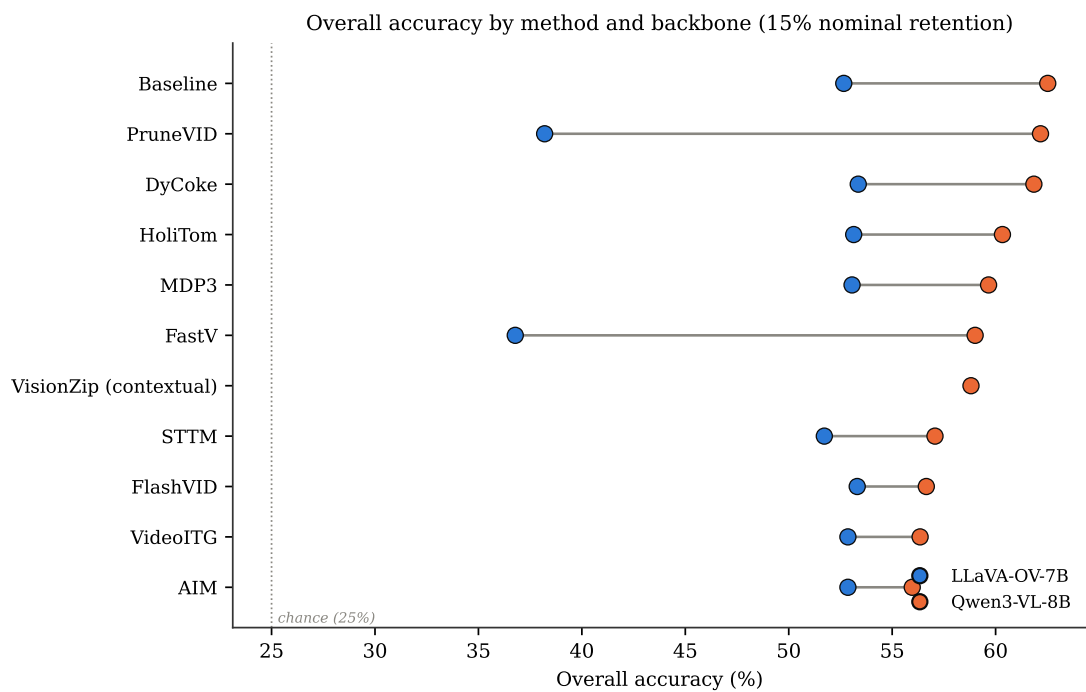


Figure 5: Overall accuracy by method and backbone at 15% nominal retention (Cleveland dot plot). Rows sorted by Qwen3-VL accuracy; the line spans each method’s two backbone results.

5 Per-method settings

Table 4: Per-method mechanism, key parameters (as run in this report), and measured visual-token budget. Budgets are read from each method’s execution log on Qwen3-VL-8B (11,664 visual input tokens at 32 frames), not from its CLI flag — nominal 15% describes the configured knob, not necessarily what is actually retained. Frame-selection methods have no token budget to measure; they instead reduce the number of frames fed to the model.

Method	Mechanism	Key parameters (this report)	Measured budget
HoliTom	pre-LLM segment retain + in-LLM merge	RETAIN_RATIO=0.15 T=0.80 HOLITOM_k=18 HOLITOM_r=0.5	7.5% (875 tok.)
STTM	quadtree spatio-temporal merge	LLaVA-OV: thresh=0.85 temporal=0.65 root=1. Qwen3-VL: thresh=0.85 temporal=-1.0 root=0 (no published config for this backbone)	9.0% (1,050 tok.)
FastV	in-LLM attention rerank, layer 2	K=2, R=0.85 (keep 15%) — outside the published K=2/R=50% range	15.0% (1,750 tok.)
FlashVID	pre-LLM temporal-segment merge	retention_ratio=0.15	15.0% (1,750 tok.)
AIM	bipartite merge + PageRank prune	2-step merge schedule compiled in; no CLI retention knob. Qwen3-VL 15%; LLaVA-OV 25% (ships two active steps)	15.0% (1,750 tok.)
VisionZip (contextual)	key-similarity merge in the vision tower	dominant=54 contextual=10 at 8 frames (native LLaVA-1.5-7B form); Qwen3-VL runs the contextual half only — no CLS token to select the dominant half	15.0% (1,750 tok.)
DyCoke	temporal merge + KV prune, layer 3	l=3 p=0.7 k=0.7 ; no single retention knob, net ~49%	49.0% (5,716 tok.)
PruneVID	DPC-KNN cluster merge, layer 10	cluster=0.50 seg=0.25 ; LLaVA-OV port has no upstream reference (upstream targets PLLaVA only)	50.0% (5,832 tok.)
MDP3	frame selection (DPP)	pool=32 select=8 — selects frames, not tokens	n/a (8 of 32 frames)
VideoITG	grounded frame selection	512 sampled → 32 selected	n/a (32 of 512 sampled)
DyTo [†]	FINCH clustering + ToMe token merge	temporal_aggregation= spatial_tome_finch_dynamic_all_frames own Vicuna-7B backbone	~3,680 tok. from 11,664 input frames

[†] DyTo does not run on either standardized backbone; listed here for completeness (see Table 6).

6 Correct/incorrect vs. baseline

Fixed = wrong under the baseline, correct under the method. *Broke* = correct under the baseline, wrong under the method. Both are McNemar discordant pairs on the 4,018 scoreable items. *Differ* counts predictions unequal to the baseline’s over all 8,052 rows (0 would mean a silent no-op — see `paper.tex` for the verification gate this guards against).

Table 5: Paired-prediction statistics against each backbone’s baseline.

Method	Differ	Fixed	Broke	χ^2	Significant
<i>LLaVA-OV-7B</i>					
DyCoke	1,031	170	142	2.34	no
FlashVID	1,610	268	242	1.23	no
HoliTom	1,838	299	280	0.56	no
MDP3	1,452	246	230	0.47	no
VideoITG	1,193	192	184	0.13	no
AIM	1,406	227	219	0.11	no
STTM	4,859	329	367	1.97	no
PruneVID-OV	4,536	473	1,054	220.30	yes
FastV	4,605	475	1,113	255.52	yes
<i>Qwen3-VL-8B</i>					
PruneVID	1,074	56	70	1.34	no
DyCoke	1,048	83	110	3.50	no
HoliTom	1,657	131	219	21.63	yes
MDP3	1,626	159	274	30.01	yes
FastV	2,029	149	290	44.65	yes
VisionZip (contextual)	2,035	191	340	41.25	yes
STTM	2,095	170	389	85.02	yes
FlashVID	2,625	178	414	93.29	yes
VideoITG	2,522	206	454	92.44	yes
AIM	2,861	194	457	105.44	yes
<i>Backbone vs. backbone</i>					
Qwen3-VL-8B vs. LLaVA-OV-7B	7,081	942	546	127.47	yes

Divergence volume does not predict damage. STTM on LLaVA-OV changes 4,859 predictions — more than FastV’s 4,605 — yet loses 0.94 points and is not significant, because its changes fall almost evenly in both directions (329 fixed, 367 broken). FastV changes a comparable number destructively (475 fixed, 1,113 broken).

7 Methods on non-standard backbones

Table 6: Runs on backbones other than the two standardized ones. Not comparable to Tables 2–3; no Δ is given because no matched baseline exists for either row.

Method	Backbone	Overall	AO	CM	LM	MR	MO	RC
VisionZip	LLaVA-1.5-7B	40.09	33.7	33.0	37.2	41.1	57.4	25.8
DyTo (recon. TW-FINCH)	Vicuna-7B	42.06	32.4	33.0	42.1	43.0	61.7	26.0

8 Notable

The backbone swap (LLaVA-OV-7B $\rightarrow$ Qwen3-VL-8B) moves accuracy by 9.86 points — more than any token-reduction method moves it in either direction, on either backbone, at any retention level in Table 1. Method rankings do not transfer across backbones: PruneVID is the worst LLaVA-OV result (38.20%) and the best Qwen3-VL one (−0.35 vs. baseline); Figure 5 makes this visible directly — several lines cross. Two methods, FastV and PruneVID-OV, collapse specifically on LLaVA-OV

(37–38%, more than 14 points below baseline and significant by McNemar’s test), while every other LLaVA-OV method is statistically indistinguishable from the backbone. On Qwen3-VL the pattern inverts: every method loses accuracy and eight of ten losses are significant, yet none approaches the LLaVA-OV collapse in size — the largest Qwen3-VL loss (AIM, -6.55) is still smaller than the smallest LLaVA-OV collapse. Tripling the retention budget from 10% to 25% moves no LLaVA-OV method more than 0.85 points, but moves FlashVID 4.63 points and FastV 3.68 on Qwen3-VL (Table 1, Figure 4) — roughly a five-fold difference in retention sensitivity between the two backbones at the same nominal settings.